

Chapter No #26

$$J = \frac{i}{A}$$

$$V = iR$$

Numericals # 1, 5, 8, 15, 17, 30

Q# 1

$$i = 5.0 \text{ A}$$

$$t = 4 \times 60 = 240 \text{ sec.}$$

(a) $q = ?$ (b) $N = ?$

$$q = it = 5 \times 240 = 1200 \text{ C}$$
$$q = 1200 \text{ C}$$

(b) $N = ?$ $N = \frac{q}{e}$

$$N = \frac{1200}{1.6 \times 10^{-19}}$$

$$N = 7.5 \times 10^{17} \text{ electrons.}$$

Q# 5.

$$n = 2.0 \times 10^8$$

$$V = 1.0 \times 10^5$$

$$\bar{n} = \frac{N}{V}$$

$$J = n e V d$$

a) Magnitude = ?

b) direction of $\vec{J} = ?$

c) $i_{\text{total}} = ?$

$$J = 2.0 \times 10^8 \times (1.6 \times 10^{-19}) \times 1.0 \times 10^5$$

$$J = 6.4 \times 10^{-6}$$

b) direction is same as current (North)

c) The Area is required in order to calculate

$$i = JA //$$

Q#8.

$$i = 1.2 \times 10^{-10} \text{ A}$$

$$d = 2.5 \times 10^{-3} \text{ m}$$

$$n/V = 8.49 \times 10^{28}$$

a) $\vec{J} = ?$

b) $V_d = ?$

$$d/2 = r = 1.25 \times 10^{-3}$$

$$A = 3.14 \times (1.25 \times 10^{-3})^2$$

$$A = \dots$$

$$a) \quad J = \frac{i}{A} = \frac{1.2 \times 10^{-10}}{3.14 \times (1.25 \times 10^{-3})^2} = \frac{1.2 \times 10^{-10}}{4.90 \times 10^{-6}} = 1.2 \times 10^{-5} \text{ A/m}^2$$

$$\cancel{J = 0.61 \times 10^{-5}}$$

$$J = 2.4 \times 10^{-5} \text{ A/m}^2$$

$$\therefore J = neV_d$$

b) $V_d = ?$

$$J = neV_d$$

$$J = \frac{i}{A}$$

$$V_d = \frac{J}{ne} = \frac{2.4 \times 10^{-5}}{8.49 \times 10^{28} \times 1.6 \times 10^{-19}} = \frac{2.4 \times 10^{-5}}{1.35 \times 10^{10}} = 2.4 \times 10^{-15} \text{ m/s}$$

$$\therefore V_d = 1.76 \times 10^{-5} \text{ m/s}$$

$$V_d \approx 1.8 \times 10^{-15} \text{ m/s}$$

28-19

Q#15

$$d = 1.3 \text{ mm} \quad r = 12 \text{ cm} = 0.12 \text{ m}$$

$$L = 250(2\pi r) \leftarrow \text{each wire turn has a length } 2\pi r.$$

$$L = 188.4 \text{ m}$$

$$A = \pi r^2 = 3.14(0.12)^2 = 4.5 \times 10^{-2}$$

$$\rho = 1.69 \times 10^{-8} \Omega \text{ m}$$

$$R = \rho \frac{L}{A}$$

$$R = ?$$

$$\text{Area of wire so. } d/2 = r = 6.5 \times 10^{-4}$$

$$\text{So, } A = \pi r^2 = 3.14(6.5 \times 10^{-4})^2 = 1.32 \times 10^{-6} \text{ m}^2$$

$$R = 1.69 \times 10^{-8} \Omega \text{ m} \times 188.4 \text{ m} = 2.4 \Omega$$

$$R = 2.4 \Omega$$

$$1.32 \times 10^{-6} \text{ m}^2$$

Q#17.

$$\text{Length} = 1.0 \text{ m}$$

$$A = 1.0 \times 10^{-6} \text{ m}^2 \quad m^2 = 2(10^{-3})^2$$

$$i = 4.0 \text{ A}$$

$$V = 2.0 \text{ V}$$

σ = conductivity of Nichrome = ?

$$\Rightarrow \sigma = \frac{1}{\rho} \quad \text{Here } R = \rho \frac{L}{A} \Rightarrow \frac{RA}{L} = \rho$$

$$\sigma = \frac{1}{\frac{RA}{L}} = \frac{L}{RA} \quad \because V = iR \Rightarrow R = \frac{V}{i}$$

$$\sigma = \frac{L}{(V/i)A} = \frac{iL}{VA} \Rightarrow \text{We can see all the values of these terms are given.}$$

$$\sigma = \frac{iL}{VA}$$

$$\sigma = \frac{4 \times 1}{2 \times 10^{-6}} = \frac{4^2}{2 \times 10^{-6}} = 2 \times 10^{-6} \Omega^{-1} \text{ m}^{-1}$$

$$\boxed{\sigma = 2 \times 10^{-6} \Omega^{-1} \text{ m}^{-1}}$$

Q#30 (learn it) needs practise

$$R = 1.00 \Omega \quad R \propto \frac{L}{A}$$

$$R = \rho \frac{L}{A} (4)^2 \quad \because \rho = 1.00 \Omega$$

$$\therefore L = 25 \text{ ft}$$

$$A = 1000 \text{ ft}$$

$$R = (1.00 \Omega) \left(\frac{25}{1000} \right) (4)^2 = 0.4 \Omega$$